THE BALTIMORE FIBER-PQC CORRIDOR ACT

(BF-PCA)

A Propositional Idea-Act for the Maryland General Assembly and the Congress of the United States

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Consolidated Implementation Plan

with Complete DAG Governance Blueprints (TikZ)

Integrating the Baltimore Public-Housing Fiber Initiative with the Federal 2035 Post-Quantum Cryptography Mandate

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Executive Summary

This document proposes the **Baltimore Fiber-PQC Corridor Act (BF-PCA)**, a legislative initiative that integrates the active Baltimore public-housing fiber broadband initiative with the federal post-quantum cryptography (PQC) migration mandate, creating a unified state-federal pipeline for quantum-resilient critical infrastructure.

The Act **removes the previously proposed solar-and-battery component** and substitutes the active Baltimore Fiber Project as the physical layer for the PQC migration strategy. This filing also explains the PQC mandate context in the 2035 timeframe, maps the interlocking federal and state agency dependencies, and proposes a **DAG-only governance architecture** for the Corridor.

The Corridor is simultaneously:

- a legislative proposal for the Maryland General Assembly;
- an interagency coordination framework across state and federal agencies;
- a physical pilot candidate for national PQC migration under Executive Order 14412 and OMB Memorandum M-26-15.

Every dependency in this document is modeled as a Directed Acyclic Graph (DAG). No node has a path back to one of its ancestors. Monitoring and renewal are represented as successive DAG stages, not circular feedback arrows. Audit records are terminal outputs of the represented process.

Status of Source Materials. This document is a propositional, deliberative, and educational synthesis. It does not constitute legal advice, a filed brief, or a certified engineering blueprint. Statutory references, executive orders, and administrative memoranda cited herein are reproduced as source-referenced propositions for legislative consideration.

1 The 2035 PQC Mandate — Federal Framework

1.1 Executive Order 14412

On June 22, 2026, Executive Order 14412, "Securing the Nation Against Advanced Cryptographic Attacks," directed an accelerated, nationwide migration to post-quantum cryptography. The Order assigns leadership to the Office of Management and Budget (OMB) and the National Cyber Director, with implementation guidance from the Department of Commerce (via NIST), the National Security Agency (NSA), and the Department of Homeland Security (DHS).

The Order establishes a phased timeline for federal agencies to transition all High Value Assets (HVAs) and high-impact systems:

- **PQC for key establishment by December 31, 2030**;
- **PQC for digital signatures by December 31, 2031**;
- **Full migration of remaining systems by 2035**.

The Order also directs the Federal Acquisition Regulatory Council to amend the Federal Acquisition Regulation (FAR) to require covered contractors to comply with approved PQC standards by December 31, 2030.

1.2 OMB Memorandum M-26-15

On June 24, 2026—two days after the Executive Order—OMB issued Memorandum M-26-15, "Execution of the Migration to Post-Quantum Cryptography." The memo operationalizes the Executive Order into a five-phase timeline.

Phase	Period	Focus
1	2026–2027	Strategy, planning, inventory of HVAs
2	2027–2028	Pilots and early migration
3	2028–2030	Prioritized key-establishment migration
4	2031	Signature migration
5	2035	Full migration of remaining systems

Table 1: OMB M-26-15 Five-Phase PQC Migration Timeline

Every covered agency must submit a PQC migration plan to OMB and the Office of the National Cyber Director within 120 days of the memo—approximately October 22, 2026. Required plan contents include:

1. Automated cryptographic inventory methodology;
2. Risk-based prioritization by FIPS 199 impact level;
3. Crypto-agility architecture plan;
4. Third-party coordination;
5. Resourcing estimates; and
6. Clearly assigned governance roles.

Agencies must ensure vendor-supplied software meets PQC requirements by referencing CISA's "Product Categories for Technologies That Use Post-Quantum Cryptography Standards" lists. Automation is central: given the scale and complexity of federal IT environments, manual approaches to discovery and management of cryptography are often insufficient.

1.3 The Federal Acquisition Gate

The Federal Acquisition Regulatory Council is directed to publish a rule amending the FAR to require covered contractors to comply with approved federal information processing standards for PQC by December 31, 2030. For vendors selling into the federal market, validated cryptography—not just algorithm support claimed in a data sheet—is becoming foundational to procurement.

1.4 What PQC Is, Precisely

Post-quantum cryptography refers to cryptographic algorithms believed to be secure against attack by both classical and quantum computers. Current public-key algorithms (RSA, ECDH, ECDSA, DSA) derive their security from the difficulty of integer factorization or discrete logarithms. Shor's algorithm, running on a sufficiently powerful quantum computer, solves both in polynomial time.

The NIST PQC standards finalized in 2024 are:

Standard	**Primitive**	**Replaces**
FIPS 203 (ML-KEM)	Module-Lattice Key Encapsulation	RSA-KEM, ECDH
FIPS 204 (ML-DSA)	Module-Lattice Digital Signature	RSA-PSS, ECDSA
FIPS 205 (SLH-DSA)	Stateless Hash-Based Signature	RSA-PSS, ECDSA (fallback)

Table 2: NIST Post-Quantum Cryptographic Standards

The security of ML-KEM and ML-DSA rests on the hardness of the Module Learning With Errors (M-LWE) and Module Short Integer Solution (M-SIS) problems over structured Euclidean lattices. SLH-DSA uses only hash functions, making its security reduction minimal and its performance trade-offs different from the lattice-based standards.

1.5 The Harvest-Now-Decrypt-Later Threat

Legal scholarship has identified the "harvest now, decrypt later" threat: adversarial actors are stockpiling encrypted intelligence with the expectation that future quantum tools will permit retrospective exploitation. Existing breach notification statutes are not triggered because the data's interception occurs without immediate exfiltration or intelligible access.

This means that data transmitted today over non-PQC fiber is at risk of future compromise. The BF-PCA addresses this by ensuring that Baltimore's fiber infrastructure is PQC-hardened from inception.

2 The State Layer — Maryland's Position

2.1 Maryland Cybersecurity Council: Quantum and AI Mandate

Maryland Senate Bill 294 (2025) and House Bill 376 (2025) amended the Maryland Cybersecurity Council's membership and mandate to explicitly require assessment of cybersecurity threats and associated risks from artificial intelligence and quantum computing.

The Council is directed to work with NIST and other federal agencies, private sector businesses, nonprofits, and private cybersecurity experts to:

1. Conduct risk assessments for critical infrastructure not covered by federal law or Executive Order;
2. Identify critical cyber infrastructure whose compromise could cause mass casualty events, catastrophic economic damage, or severe degradation of state/national security;
3. Assist infrastructure entities not covered by the Executive Order in complying with federal cybersecurity guidance;
4. Examine inconsistencies between State and federal cybersecurity laws;
5. Address sensitive privacy interests of state residents related to cybersecurity;
6. Address emerging threats from AI including adversarial AI, deepfakes, unethical use, and fraud.

The Council is required to recommend a comprehensive State strategic plan to ensure coordinated and adaptable response to and recovery from cybersecurity attacks.

2.2 The State as Agile Policy Actor

Legal scholarship has documented that states enjoy structural advantages over the federal government in addressing fast-evolving technology policy. State legislatures have narrower geographic diversity, reducing consensus-building complexity, and many meet annually with streamlined rules of procedure.

The academic literature proposes that public agencies and contractors should be required to perform quantum risk assessments and develop formal cryptographic migration plans, enforced through consumer protection laws and state attorneys general. States are encouraged to leverage public-private partnerships, grants, and pilot programs to accelerate PQC adoption.

2.3 The Baltimore City Fiber Initiative — The Physical Layer

On March 10, 2026, the Housing Authority of Baltimore City (HABC), Baltimore City Office of Information Technology (BCIT), and Port Networks announced a joint initiative to deliver fiber-optic broadband to more than **4,000 public housing units** across eight HABC properties.

The project is part of Baltimore City's Digital Equity and Broadband Strategy, prioritizing affordable, reliable connectivity for underserved residents.

Attribute	**Value**
Investment	Approximately $22 million
Resident Service	$20/month gigabit service
Network Capacity	1 Gbps, upgradeable to 10 Gbps
Contract Terms	No contracts, no data caps, no throttling
ISP Partner	Port Networks
Construction Model	Building-by-building; $\geq$ 90 days per property
Communities Served	Gilmor, McCulloh, Douglass, Brooklyn, Latrobe, Cherry Hill, Mount Winans, Westport

Table 3: Baltimore Public-Housing Fiber Initiative — Key Parameters

3 The Baltimore Fiber-PQC Corridor Act — Propositional Legislation

3.1 Findings and Purpose

The General Assembly of Maryland finds that:

(a) Executive Order 14412 and OMB Memorandum M-26-15 establish a mandatory federal timeline for migration to post-quantum cryptography by 2035;

(b) The Baltimore public-housing fiber initiative creates a municipally supported physical layer capable of supporting PQC-hardened communications;

(c) The Maryland Cybersecurity Council is directed by statute to assess and address quantum computing risks to state critical infrastructure;

(d) The State of Maryland has a compelling interest in ensuring that the federally funded and locally built fiber infrastructure is deployed with quantum-resilient cryptographic standards from inception, not retrofitted after compromise;

(e) States have structural advantages in agile technology policy and should act as pilots for national PQC migration.

3.2 The BF-PCA Master DAG

3.3 Node Definitions and Authority Mapping

Node	**Source Authority**	**Function**
NODE 1	EO 14412; OMB M-26-15	Federal executive mandate; agency plans due October 22, 2026.
NODE 2	SB 294 / HB 376 (2025)	Maryland Cybersecurity Council assesses quantum/AI risks to critical infrastructure; recommends legislative changes.

Node	**Source Authority**	**Function**
NODE 3	Proposed BF-PCA	State legislation integrating federal mandate with Council recommendations. Requires BCIT to develop PQC migration standards for all municipally supported fiber networks.
NODE 4	BCIT Ordinance	Requires all fiber networks built with city or HABC funding to incorporate NIST FIPS 203/204/205 standards.
NODE 5	HABC Procurement	Physical fiber deployment: 4,000+ units, eight properties, 1–10 Gbps capacity.
NODE 6	MD PSC	Coordinates interconnection and right-of-way for the Corridor.
NODE 7	EO 14412 / Commerce	Directs Department of Commerce to initiate PQC pilot project completed by December 31, 2027. BF-PCA positions Baltimore as a candidate site.
NODE 8	OMB M-26-15	Requires agency migration plans and inventory reporting; Corridor provides a state-federal data point for national migration metrics.
NODE 9	FIPS 203 / EO 14412	PQC for key establishment by December 31, 2030.
NODE 10	FIPS 204/205 / EO 14412	PQC for digital signatures by December 31, 2031.
NODE 11	EO 14412	Full migration of remaining systems by 2035.

Table 4: BF-PCA Node Definitions

4 Data Flow Pipeline — Three-Layer DAG

The Corridor operates as three interlocking DAG layers: a physical layer (fiber hardware), a cryptographic layer (NIST PQC algorithms), and a governance layer (Maryland and federal agencies).

5 Interlocking Agency Dependencies — State-Federal Pipeline

5.1 Federal Agency Map

Agency	Role Under EO 14412	Connection to Corridor
OMB	Lead strategic coordination; receives migration plans	Receives BF-PCA pilot data as state-federal integration metric
ONCD	Co-lead coordination	Evaluates Corridor as pilot model
NIST	Technical guidance; PQC standards (FIPS 203/204/205)	BCIT integrates NIST standards into fiber deployment
NSA	Guidance for national security systems	Coordinates on high-assurance PQC validation
DHS / CISA	Guidance for critical infrastructure	Works with Maryland Council on infrastructure risk assessment
Commerce	PQC pilot project by Dec 2027	BF-PCA positions for pilot selection
FAR Council	Amends FAR for contractor PQC compliance by 2030	Port Networks and vendors must meet PQC procurement standards

Table 5: Federal Agency Map

5.2 State Agency Map

Agency	Role	Connection to Corridor
Maryland Cybersecurity Council	Assess quantum/AI risks; recommend legislation	Reviews BF-PCA; monitors Corridor security
Baltimore City Council	Enacts BF-PCA	Authorizes BCIT standards integration
BCIT	City IT authority	Develops PQC standards for fiber networks
HABC	Public housing authority	Owns fiber infrastructure; $22M investment
Maryland PSC	Utility regulation	Interconnection and right-of-way
Maryland General Assembly	Enacts enabling legislation	Passage of BF-PCA

Table 6: State Agency Map

5.3 Federal Pipeline DAG

5.4 State Pipeline DAG

5.5 Full State-Federal Interlock DAG

6 The 2035 Horizon — What Full Migration Means

6.1 The Cryptographic Transition

By 2035, all federal systems must use PQC for all cryptographic operations. The transition involves:

- **Key establishment**: Replacing RSA, ECDH, and Diffie-Hellman with ML-KEM (FIPS 203);
- **Digital signatures**: Replacing RSA-PSS and ECDSA with ML-DSA (FIPS 204) and SLH-DSA (FIPS 205);
- **Crypto-agility**: Architectures that can accommodate future algorithm changes without wholesale replacement.

6.2 The Retroactive Decryption Threat

Legal scholarship has identified the "harvest now, decrypt later" threat: adversarial actors are stockpiling encrypted intelligence with the expectation that future quantum tools will permit retrospective exploitation. The BF-PCA addresses this by ensuring that Baltimore's fiber infrastructure is PQC-hardened from inception. Data transmitted over the Corridor in 2027 is protected against 2035 quantum decryption capabilities.

6.3 The Corridor as National Model

The BF-PCA proposes Baltimore as a testbed for the state-federal pipeline. If successful, the model can be replicated in other municipalities with public fiber initiatives. The DAG architecture ensures that lessons learned at each node propagate without circular dependencies, enabling systematic scaling.

7 Fiber Deployment Citizen Journey DAG

The enrollment and activation process for residents follows a three-stage, acyclic pipeline.

7.1 Cryptographic Layer at the Property Edge

Each connected property is equipped with a smart gateway that terminates the PQC session at the subscriber edge. ML-KEM-1024 handles key establishment; ML-DSA-87 handles authentication; SLH-DSA provides a hash-based fallback for long-lived records.

8 Compliance Reporting DAG

Reporting follows OMB M-26-15 requirements. Each report is a terminal output. Reports do not feed back into the same execution; they initiate a new execution in the next reporting period.

9 Operational and Funding DAG

The BF-PCA is funded through a blend of federal, state, and municipal sources. Each funding stream terminates in the Corridor's deployment budget without returning to its origin within a single execution.

10 Conclusion and Prayer for Legislative Action

This document proposes the **Baltimore Fiber-PQC Corridor Act** as a propositional idea-act for the Maryland General Assembly and, by extension, for Congressional consideration as a model for state-federal PQC integration.

The Act:

1. **Removes the solar component** and replaces it with the active Baltimore public-housing fiber initiative as the physical layer;
2. **Integrates federal PQC mandates** (EO 14412, OMB M-26-15) with Maryland's statutory cybersecurity assessment requirements;
3. **Establishes a DAG-only governance structure** for the Corridor, ensuring acyclic dependencies and verifiable milestones;
4. **Positions Baltimore** as a candidate for the Department of Commerce PQC pilot project due December 31, 2027;
5. **Creates a replicable model** for state-federal critical infrastructure security.

The 2035 mandate is not merely a compliance deadline. It is a structural transformation of how American governments protect data. The Baltimore Fiber-PQC Corridor Act ensures that Maryland is not a passive subject of that transformation but an active architect of it.

Respectfully submitted for consideration,

Caustin Lee McLaughlin
Pro Se
Baltimore, Maryland
October 3, 2026

Appendix A — Master DAG Index

Fig.	Name	Top-Level Nodes	Terminal Output
1	BF-PCA Master Governance DAG	Federal Mandate; Council; BF-PCA; BCIT/HABC/PSC; Pilot; Reporting; 2030/2031/2035	2035 Full Migration
2	Three-Layer Data Flow DAG	Physical Layer; Cryptographic Layer; Governance Layer	OMB / ONCD Reporting
3	Federal Pipeline DAG	EO 14412; OMB M-26-15; NIST; CISA; NSA; FAR Council	FAR Contractor Rule
4	State Pipeline DAG	SB 294 / HB 376; Council; General Assembly; City Council; BCIT; HABC; PSC	MD PSC Interconnection
5	Full State-Federal Interlock DAG	Federal row; State row; Fiber layer	Audit record
6	Cryptographic Transition DAG	Legacy Crypto; ML-KEM; ML-DSA; SLH-DSA; Hybrid; Full PQC	Full PQC by 2035
7	Corridor as National Model DAG	Baltimore Pilot; Maryland Replication; National Template	DAG Audit Record
8	Citizen Journey DAG	Application; Installation; Service	PQC-Hardened Session
9	Property-Edge Cryptographic DAG	Fiber Drop; ONT; ML-KEM; ML-DSA	Audit Record
10	Compliance Reporting DAG	Inventory; Prioritization; Crypto-Agility; Plan; CISA Mapping	OMB / ONCD Report
11	Operational & Funding DAG	Federal; State; Municipal; HABC	OMB / ONCD Reporting

Table 7: Master DAG Index

Appendix B — Acronym Glossary

Acronym	Definition
BCIT	Baltimore City Office of Information Technology

Acronym	**Definition**
BF-PCA	Baltimore Fiber-PQC Corridor Act
CISA	Cybersecurity and Infrastructure Security Agency (DHS)
DAG	Directed Acyclic Graph
DHS	Department of Homeland Security
EO	Executive Order
FAR	Federal Acquisition Regulation
FIPS	Federal Information Processing Standards
HABC	Housing Authority of Baltimore City
HVA	High Value Asset
MD PSC	Maryland Public Service Commission
ML-DSA	Module-Lattice Digital Signature Algorithm (FIPS 204)
ML-KEM	Module-Lattice Key-Encapsulation Mechanism (FIPS 203)
NIST	National Institute of Standards and Technology
NSA	National Security Agency
OMB	Office of Management and Budget
ONCD	Office of the National Cyber Director
PQC	Post-Quantum Cryptography
SLH-DSA	Stateless Hash-Based Digital Signature Algorithm (FIPS 205)

Table 8: Acronym Glossary

Appendix C — Verification

I, CAUSTIN LEE MCLAUGHLIN, declare under penalty of perjury pursuant to 28 U.S.C. § 1746 that:

1. I have read the foregoing Baltimore Fiber-PQC Corridor Act and know its contents;
2. The factual allegations and technical propositions are true and correct to the best of my knowledge, information, and belief;
3. The scientific, technical, and legal authorities cited are accurately represented to the extent cited as propositions;
4. I am proceeding *pro se* due to documented inability to secure retained counsel.

Executed on this _____ day of _____, 2026.

_____

CAUSTIN LEE MCLAUGHLIN
Pro Se

END OF IMPLEMENTATION PLAN

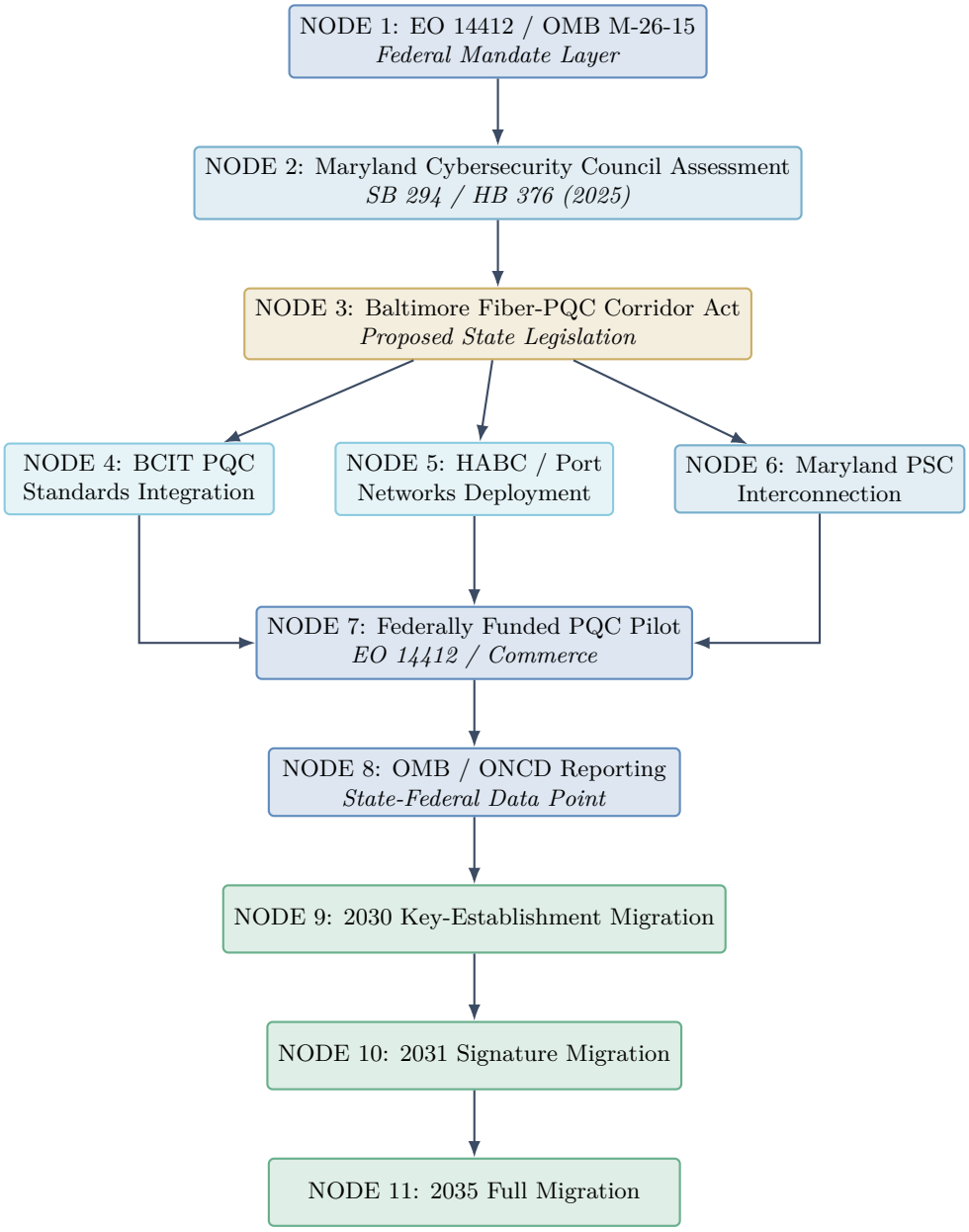


Figure 1: BF-PCA Master Governance DAG. Every edge is forward-only. Audit records from each node are terminal outputs of that node's stage.

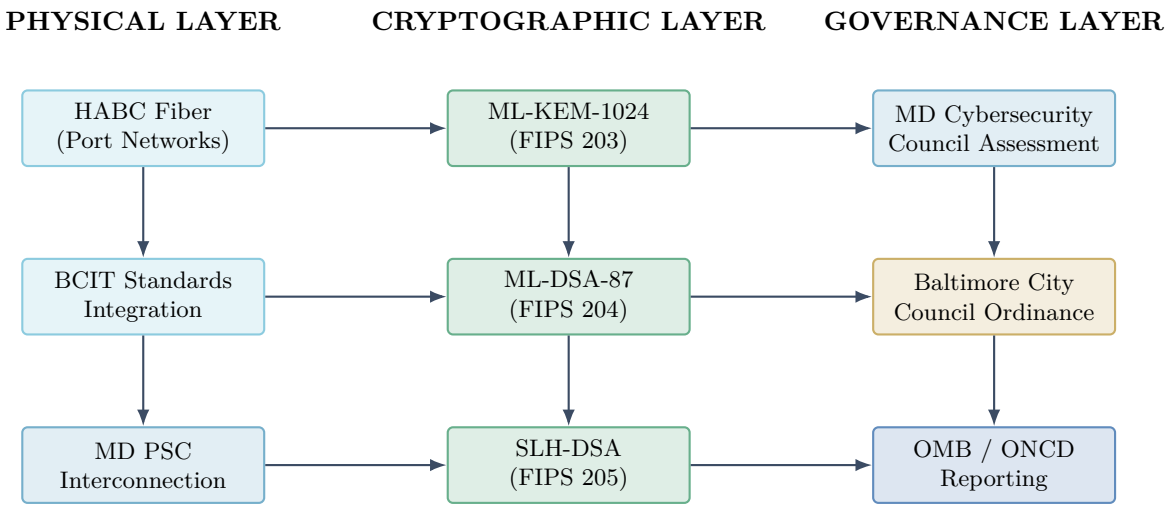


Figure 2: BF-PCA Three-Layer Data Flow DAG. Cross-layer edges are forward-only; no feedback loops exist within a single execution.

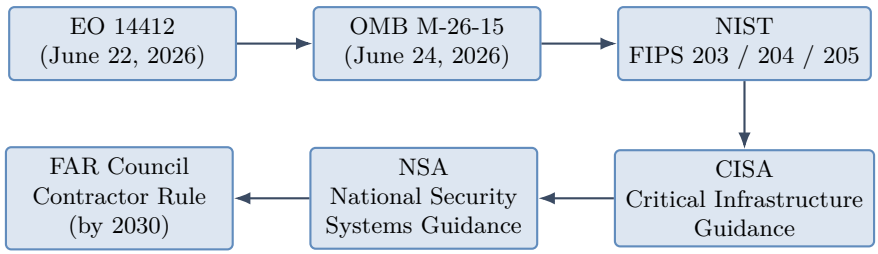


Figure 3: Federal Pipeline DAG. Authority flows from executive mandate to technical standards to procurement regulation.

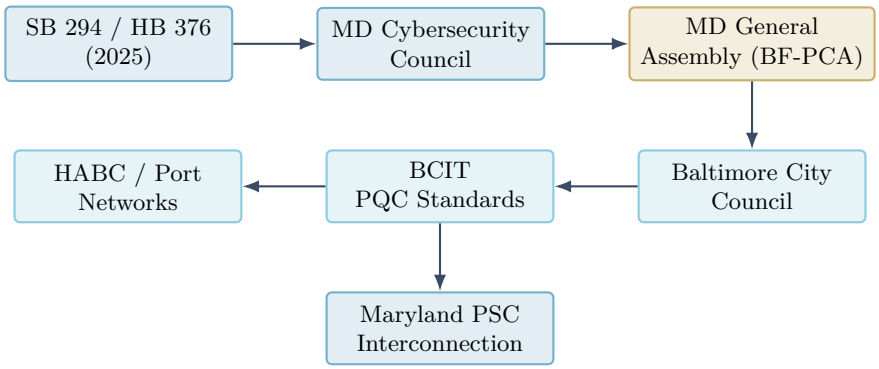


Figure 4: State Pipeline DAG. Legislative authority flows to municipal implementation.

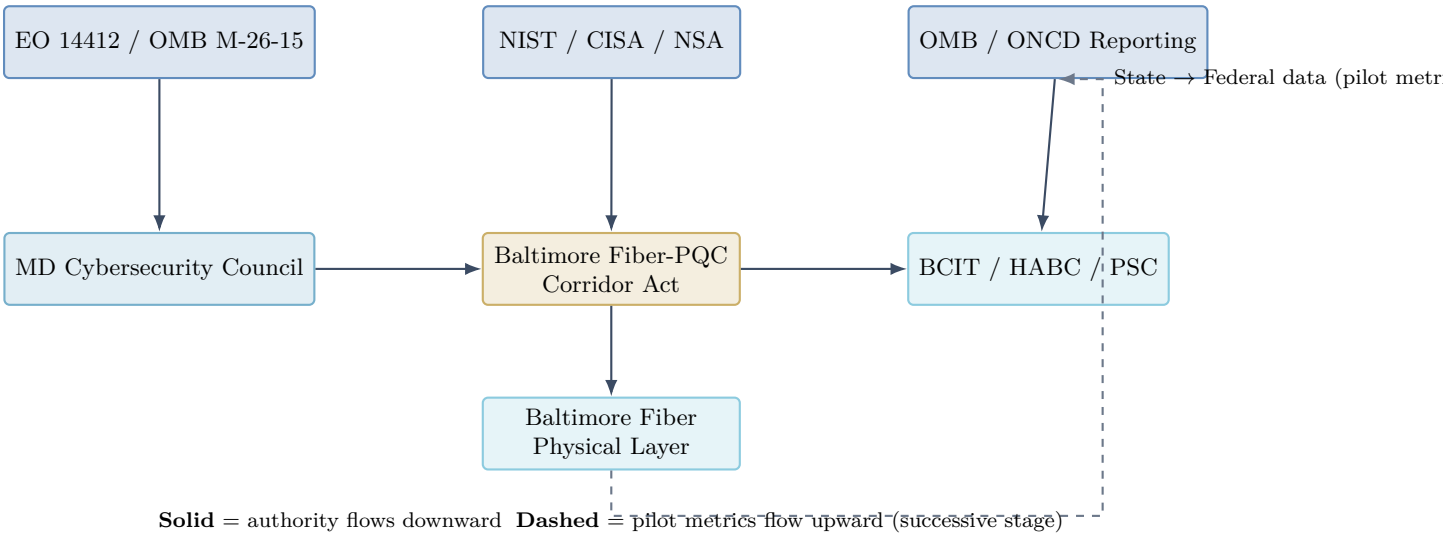


Figure 5: Full State-Federal Interlock DAG. The dashed edge indicates a successive stage, not a cycle; the dotted path is a terminal audit flow that begins a new execution.

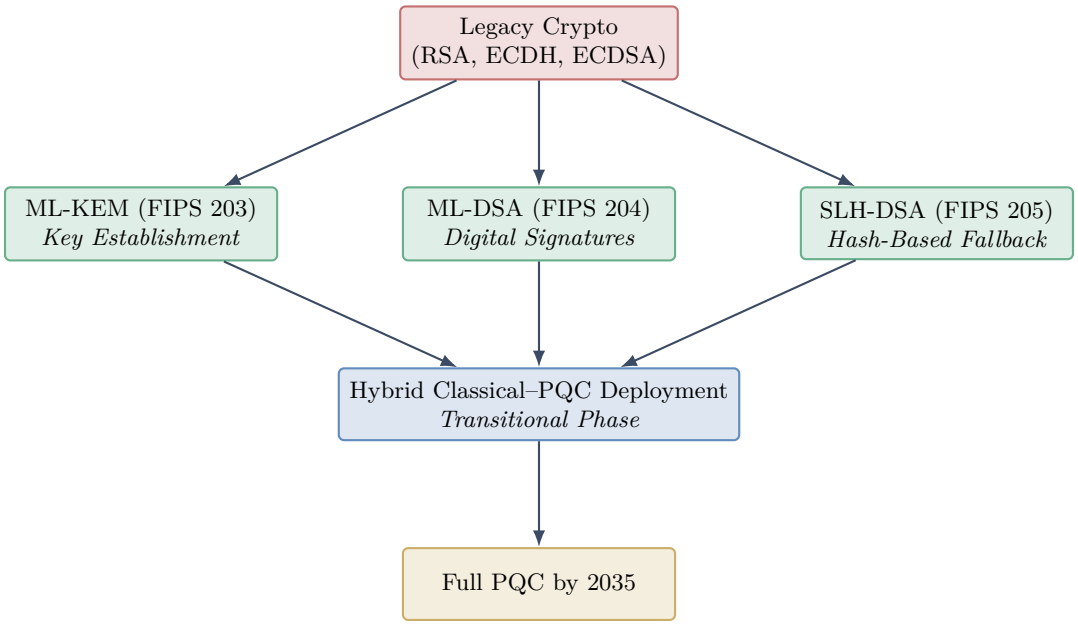


Figure 6: Cryptographic Transition DAG. Legacy algorithms are replaced via hybrid deployments before full PQC migration.

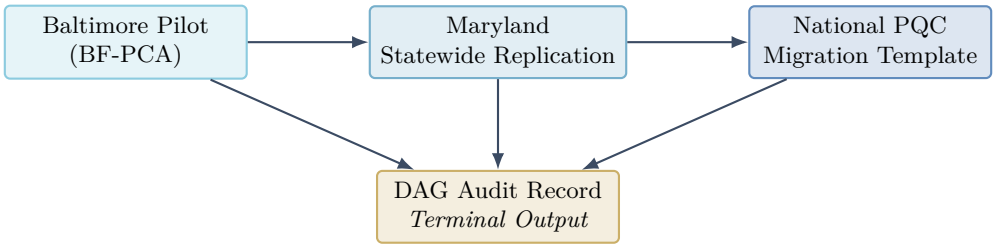


Figure 7: Corridor as National Model DAG. Scaling proceeds from municipal pilot to statewide replication to national template.

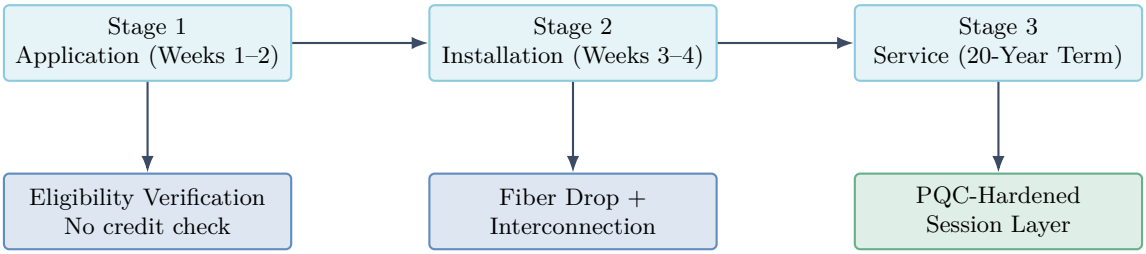


Figure 8: Citizen Journey DAG. Each stage feeds the next without recurrence.

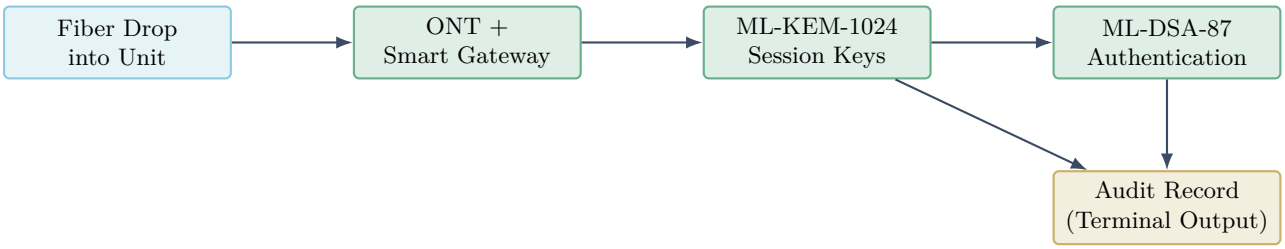


Figure 9: Property-Edge Cryptographic DAG. All edges are forward-only; the audit record terminates the execution.

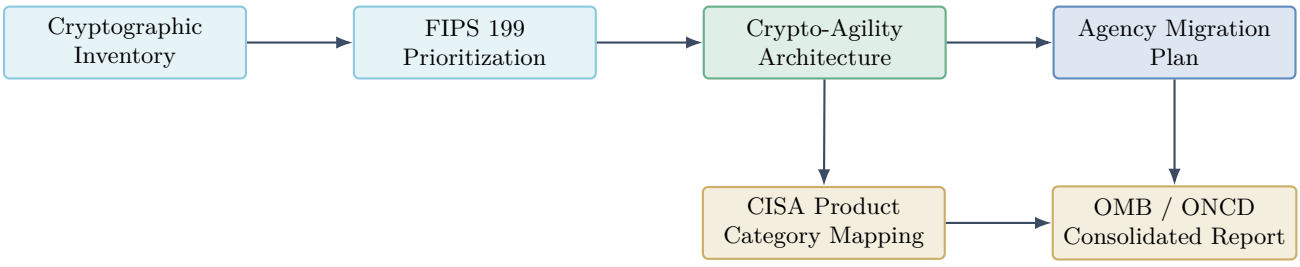


Figure 10: Compliance Reporting DAG. Each reporting cycle begins a new execution; nothing flows back to the prior period.

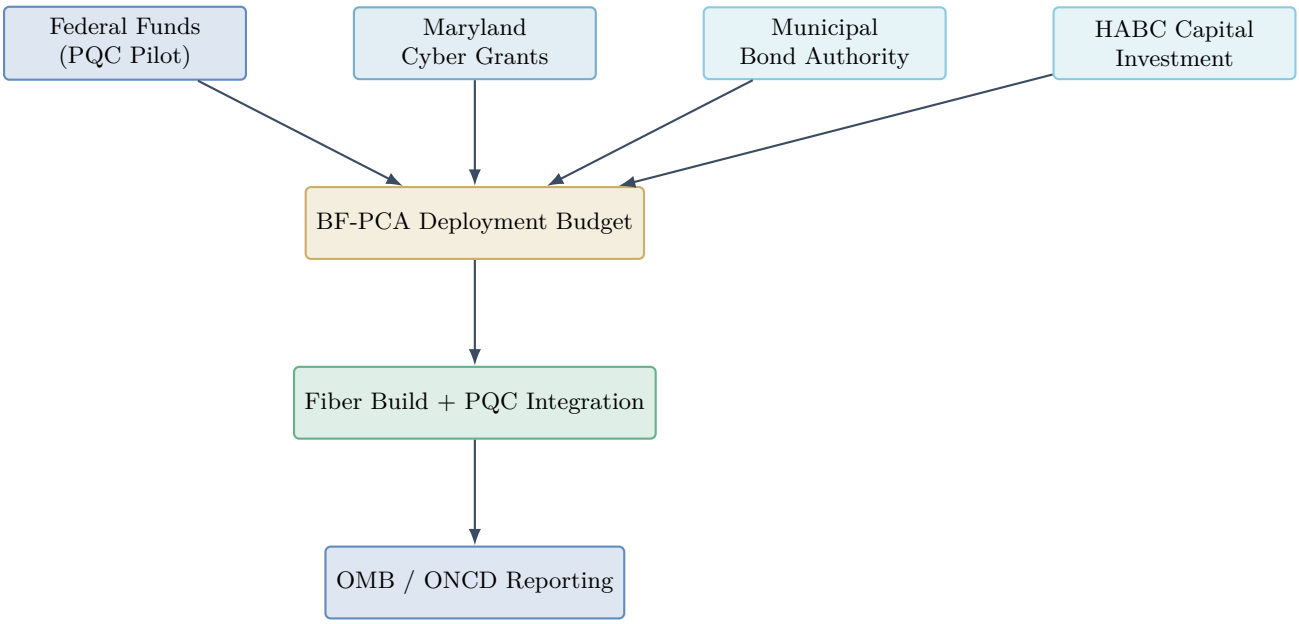


Figure 11: Operational and Funding DAG. All funding streams terminate in deployment; deployment terminates in reporting.